

Airline Data Management and Analysis Using Power BI

Task 1 : Data Preparation and Cleaning

I imported the Flight Information, Passenger Information, and Ticket Information datasets into Power Query Editor.

I removed duplicate entries, handled missing values, and formatted columns properly to ensure clean and usable data for analysis.

Passenger information table:

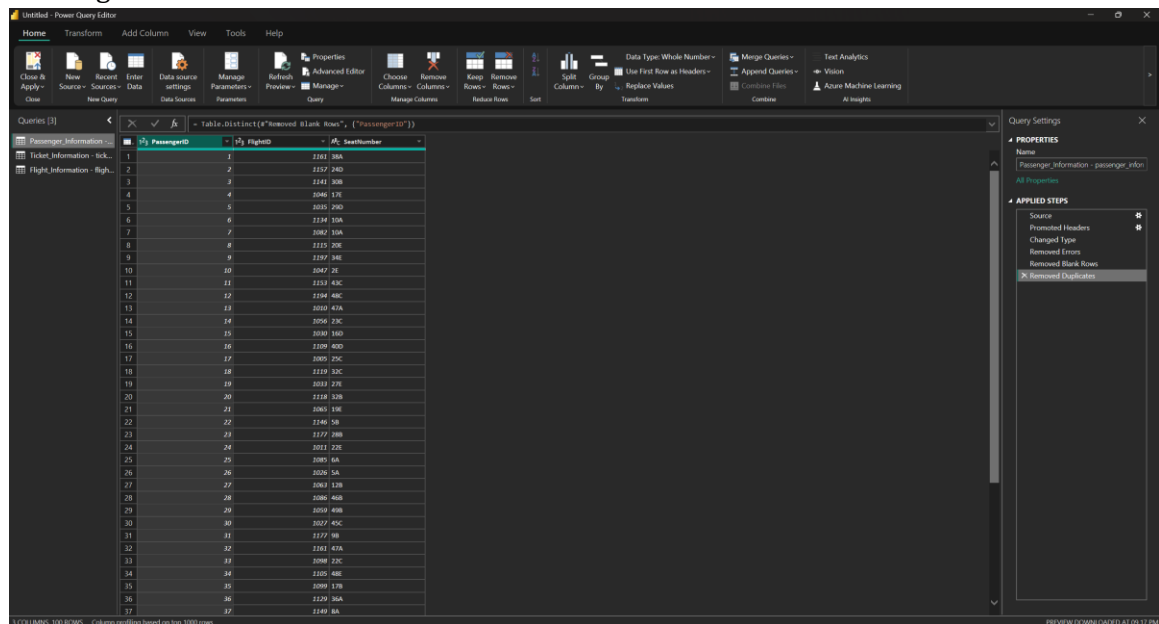


Table: Table.Distinct("removed blank rows", {"PassengerID"})

PassengerID	FlightID	SeatNumber
1	1	1181 38A
2	2	1157 24D
3	3	1141 30B
4	4	1046 17E
5	5	1035 29D
6	6	1134 10A
7	7	1082 10A
8	8	1115 20E
9	9	1157 14E
10	10	1047 2E
11	11	1113 43C
12	12	1104 48C
13	13	1010 47A
14	14	1050 12C
15	15	1030 16D
16	16	1109 40D
17	17	1005 25C
18	18	1119 82C
19	19	1011 27E
20	20	1118 32B
21	21	1065 19E
22	22	1140 58
23	23	1177 19B
24	24	1011 22E
25	25	1085 6A
26	26	1026 5A
27	27	1083 17B
28	28	1046 40B
29	29	1019 49B
30	30	1027 45C
31	31	1177 9B
32	32	1143 47A
33	33	1088 12C
34	34	1105 48E
35	35	1089 17B
36	36	1129 36A
37	37	1149 6A

Ticket information table:

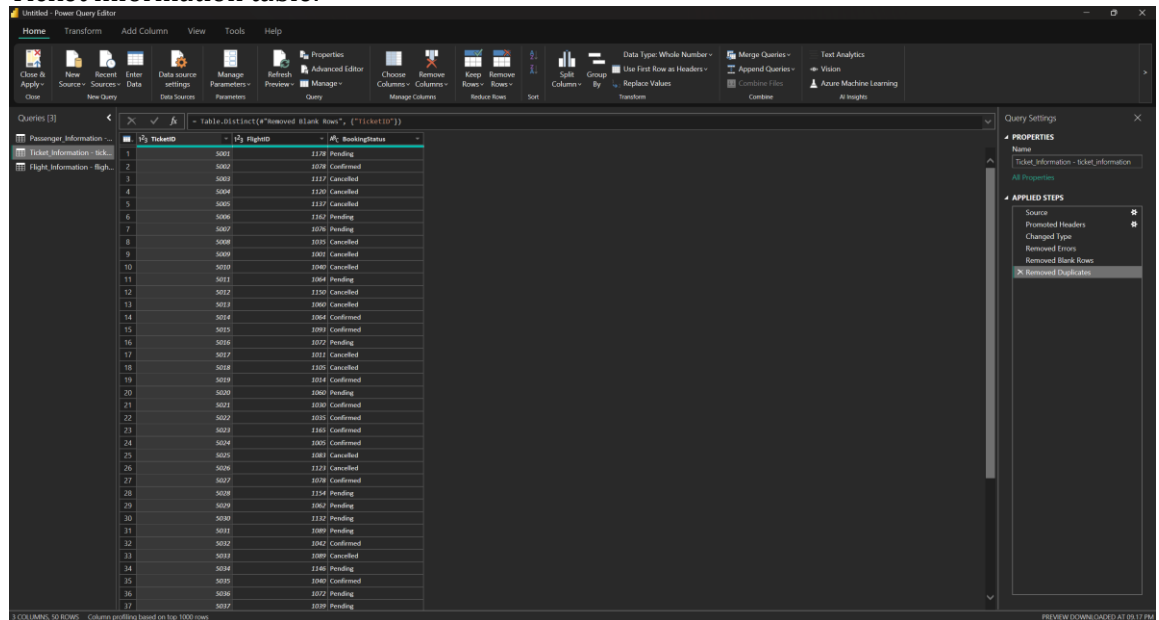


Table: Table.Distinct("removed blank rows", {"TicketID"})

TicketID	FlightID	BookingStatus
1	1178	Pending
2	1078	Confirmed
3	1117	Cancelled
4	1120	Cancelled
5	1117	Cancelled
6	1162	Pending
7	1078	Pending
8	1079	Cancelled
9	1001	Cancelled
10	1040	Cancelled
11	1084	Pending
12	1120	Cancelled
13	1090	Cancelled
14	1064	Confirmed
15	1093	Confirmed
16	1072	Pending
17	1011	Cancelled
18	1105	Cancelled
19	1014	Confirmed
20	1060	Pending
21	1030	Confirmed
22	1010	Confirmed
23	1161	Confirmed
24	1005	Confirmed
25	1081	Cancelled
26	1121	Cancelled
27	1078	Confirmed
28	1114	Pending
29	1062	Pending
30	1132	Pending
31	1089	Pending
32	1042	Confirmed
33	1089	Cancelled
34	1146	Pending
35	1080	Confirmed
36	1072	Pending
37	1019	Pending

Flight information table:

HomeTransformAdd ColumnViewToolsHelp

Close & Apply

New Source

Recent Sources

Enter Data

Data source settings

Data Sources

Manage Parameters

Refresh

Advanced Editor

Preview

Manage

Query

Choose Columns

Remove Columns

Keep Rows

Remove Rows

Reduce Rows

Sort

Split Column

Group By

Use First Row as Headers

Replace Values

Transform

Data Type: Whole Number

Merge Queries

Append Queries

Combine Files

Continue

Test Analytics

Visuals

Azure Machine Learning

AI Insights

Queries (3)

Flight

Table.Distinct("removed blank rows")

Passenger Information - tick...

Ticket Information - tick...

Flight Information - flight...

	FlightID	FlightNumber	Airline	Destination	Status
1	2003	FL1302	Airline D	Houston	On Time
2	2002	FL1435	Airline B	Chicago	Cancelled
3	2003	FL1860	Airline A	New York	Cancelled
4	2004	FL1270	Airline C	Chicago	Delayed
5	2005	FL1306	Airline C	New York	Delayed
6	2006	FL1071	Airline A	Phoenix	On Time
7	2007	FL1700	Airline C	Los Angeles	Cancelled
8	2008	FL1020	Airline C	Los Angeles	Delayed
9	2009	FL1014	Airline A	Los Angeles	Cancelled
10	2010	FL1121	Airline D	Chicago	Cancelled
11	2011	FL1466	Airline A	Phoenix	On Time
12	2012	FL1214	Airline D	New York	Delayed
13	2013	FL1330	Airline C	Houston	On Time
14	2014	FL1408	Airline C	New York	Delayed
15	2015	FL1087	Airline C	Houston	Delayed
16	2016	FL1372	Airline B	New York	Delayed
17	2017	FL1099	Airline D	Phoenix	Delayed
18	2018	FL1871	Airline B	Houston	Cancelled
19	2019	FL1663	Airline B	Chicago	Cancelled
20	2020	FL1130	Airline A	New York	On Time
21	2021	FL1661	Airline B	New York	Cancelled
22	2022	FL1308	Airline A	Houston	Delayed
23	2023	FL1409	Airline A	Chicago	On Time
24	2024	FL1361	Airline B	Chicago	Delayed
25	2025	FL1491	Airline D	Phoenix	On Time
26	2026	FL1413	Airline D	Chicago	Cancelled
27	2027	FL1805	Airline D	Chicago	On Time
28	2028	FL1265	Airline D	Chicago	On Time
29	2029	FL1181	Airline D	Los Angeles	On Time
30	2030	FL1955	Airline B	Phoenix	On Time
31	2031	FL1276	Airline B	New York	On Time
32	2032	FL1360	Airline C	Houston	Delayed
33	2033	FL1409	Airline D	New York	On Time
34	2034	FL1313	Airline B	Phoenix	On Time
35	2035	FL1021	Airline C	Houston	Cancelled
36	2036	FL1252	Airline D	Phoenix	On Time
37	2037	FL1747	Airline A	Chicago	Cancelled

5 COLUMNS, 200 ROWS

Column profiling based on top 1000 rows

Query Settings

PROPERTIES

Name

Flight Information - flight_information

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

Removed Errors

Removed Blank Rows

Removed Duplicates

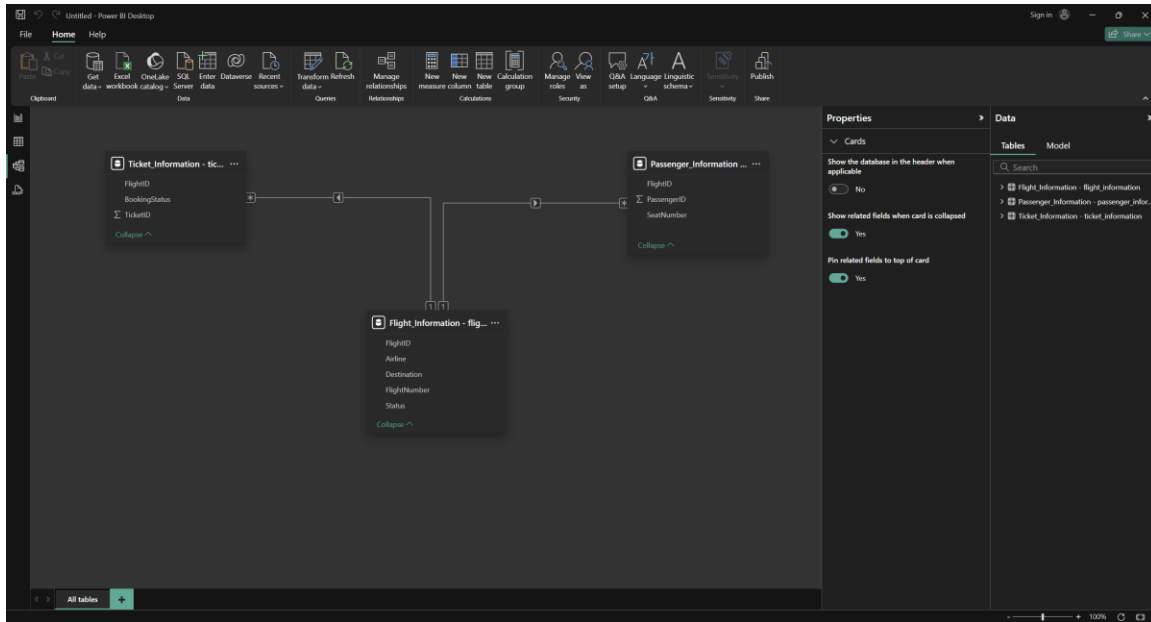
PREVIEW DOWNLOADED AT 09:17 PM

Insight: Cleaning the data improved its reliability and ensured that all further analysis would be accurate and error-free.

Task 2: Data Modeling

I created relationships between the datasets using FlightID as the primary key. I configured the correct cardinality and ensured proper cross-filtering between the tables to create a robust data model.

Relationship model view:



Insight: Establishing accurate relationships made it easy to connect flight, passenger, and ticket information for meaningful cross-analysis.

Task 3: Enhanced Data Insights

I added a conditional column classifying flights as "Best" or "To Be Improved" based on their status.

I also used the "Column from Examples" feature to extract numeric parts of the FlightNumber for cleaner visualizations and easier data manipulation.

Unfiled - Power Query Editor

HomeTransformAdd ColumnViewToolsHelp

Columns from Custom Inverse Custom Conditional Column Examples - Function Duplicate Column

Format - Extract - Paste - General

Statistics - Standard - Scientific - Rounding - Information -

Date - Time - Duration - From Date & Time

Test Analytics - Vision - Azure Machine Learning - Alerts

Columns from Custom Inverse Custom Conditional Column Examples - Function Duplicate Column

Format - Extract - Paste - General

Statistics - Standard - Scientific - Rounding - Information -

Date - Time - Duration - From Date & Time

Test Analytics - Vision - Azure Machine Learning - Alerts

Queries [1]

Table.RenameColumns(*"Inserted Text After Delimiter",{"FlightNumber","Airline number"})

	FlightID	Airline number	Airline	Destination	Status	Flight_rating	Flight_Number
1	10007	FL1002	Airline D	Houston	On Time	Best	1002
2	10002	FL1435	Airline B	Chicago	On Time	Best	1435
3	10003	FL1960	Airline A	New York	Cancelled	To be improved	1960
4	1004	FL1270	Airline C	Chicago	Delayed	To be improved	1270
5	1005	FL1106	Airline C	New York	Delayed	To be improved	1106
6	1006	FL1071	Airline A	Phoenix	On Time	Best	1071
7	1007	FL1700	Airline C	Los Angeles	Cancelled	To be improved	1700
8	1008	FL1020	Airline C	Los Angeles	Delayed	To be improved	1020
9	1009	FL1214	Airline A	Los Angeles	Cancelled	To be improved	1214
10	1010	FL1121	Airline D	Chicago	Cancelled	To be improved	1121
11	1011	FL1466	Airline A	Phoenix	On Time	Best	1466
12	1012	FL1214	Airline D	New York	Delayed	To be improved	1214
13	1013	FL1130	Airline C	Houston	On Time	Best	1130
14	1014	FL1458	Airline C	New York	Delayed	To be improved	1458
15	1015	FL1087	Airline C	Houston	Delayed	To be improved	1087
16	1016	FL1372	Airline B	New York	Delayed	To be improved	1372
17	1017	FL1099	Airline D	Phoenix	Delayed	To be improved	1099
18	1018	FL1871	Airline B	Houston	Delayed	To be improved	1871
19	1019	FL1063	Airline B	Chicago	Cancelled	To be improved	1063
20	1020	FL1130	Airline A	New York	On Time	Best	1130
21	1021	FL1861	Airline B	New York	Cancelled	To be improved	1861
22	1022	FL1308	Airline A	Houston	Delayed	To be improved	1308
23	1023	FL1769	Airline A	Chicago	On Time	Best	1769
24	1024	FL1242	Airline B	Chicago	Delayed	To be improved	1242
25	1025	FL1491	Airline D	Phoenix	On Time	Best	1491
26	1026	FL1413	Airline D	Chicago	Cancelled	To be improved	1413
27	1027	FL1805	Airline D	Chicago	On Time	Best	1805
28	1028	FL1185	Airline D	Chicago	On Time	Best	1185
29	1029	FL1131	Airline D	Los Angeles	On Time	Best	1131
30	1030	FL1905	Airline B	Phoenix	On Time	Best	1905
31	1031	FL1276	Airline B	New York	On Time	Best	1276
32	1032	FL1160	Airline C	Houston	Delayed	To be improved	1160
33	1033	FL1404	Airline D	New York	On Time	Best	1404
34	1034	FL1313	Airline B	Phoenix	On Time	Best	1313
35	1035	FL1021	Airline C	Houston	Cancelled	To be improved	1021
36	1036	FL1252	Airline D	Phoenix	On Time	Best	1252
37	1037	FL1747	Airline A	Chicago	Cancelled	To be improved	1747

Query Settings

PROPERTIES

NameFlight Information - Right informationAll Properties

APPLIED STEPSSourcePromoted HeadersChanged TypeRemoved ErrorsRemoved Blank RowsRemoved DuplicatesAdded Conditional ColumnRenamed ColumnsInserted Text After DelimiterRenamed Column1

7 COLUMNS, 200 ROWS

Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 09:40

Insight: Creating new columns helped classify and organize the flights, making trend analysis much more straightforward.

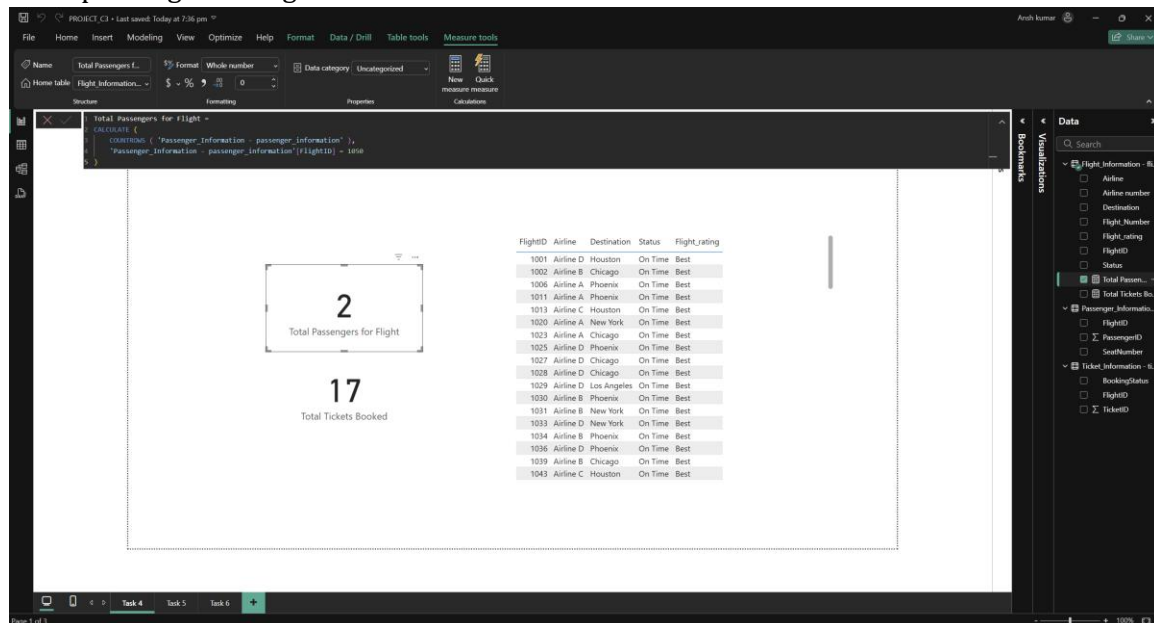
Task 4: Calculations Using DAX

I wrote DAX measures to calculate:

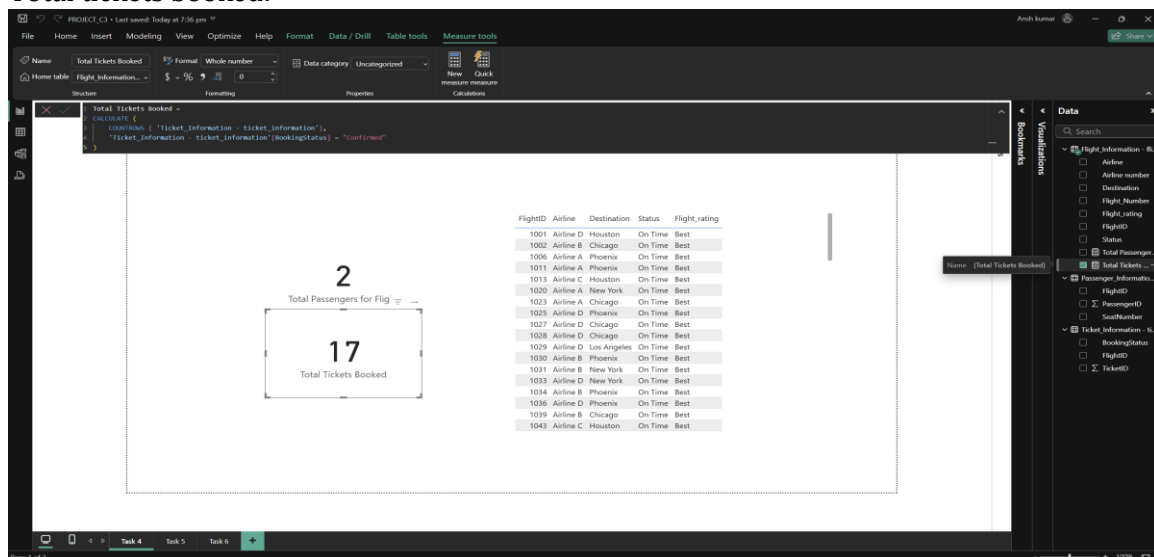
- Total passengers per flight
- Total tickets booked
- A filtered table showing only the "Best" flights

These calculations allowed for deeper exploration into operational efficiency and booking trends.

Total passenger for flights:



Total tickets booked:



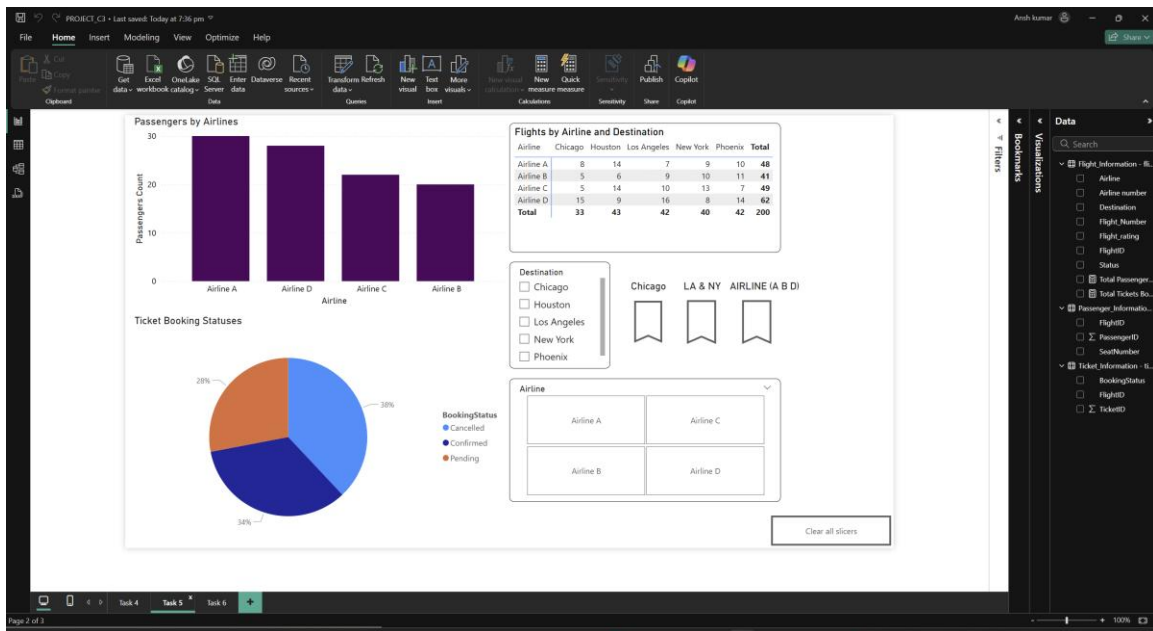
Insight: Using DAX unlocked dynamic calculations that provided sharper insights into passenger load and flight performance.

Task 5: Visualization and Interactive Features

I created various visuals, including:

- Passenger count by airline
- Ticket booking status overview
- Flights categorized by airline and destination

Interactive features such as slicers for Destination and Airline were added, along with dedicated airline-specific features for deeper individual analysis.

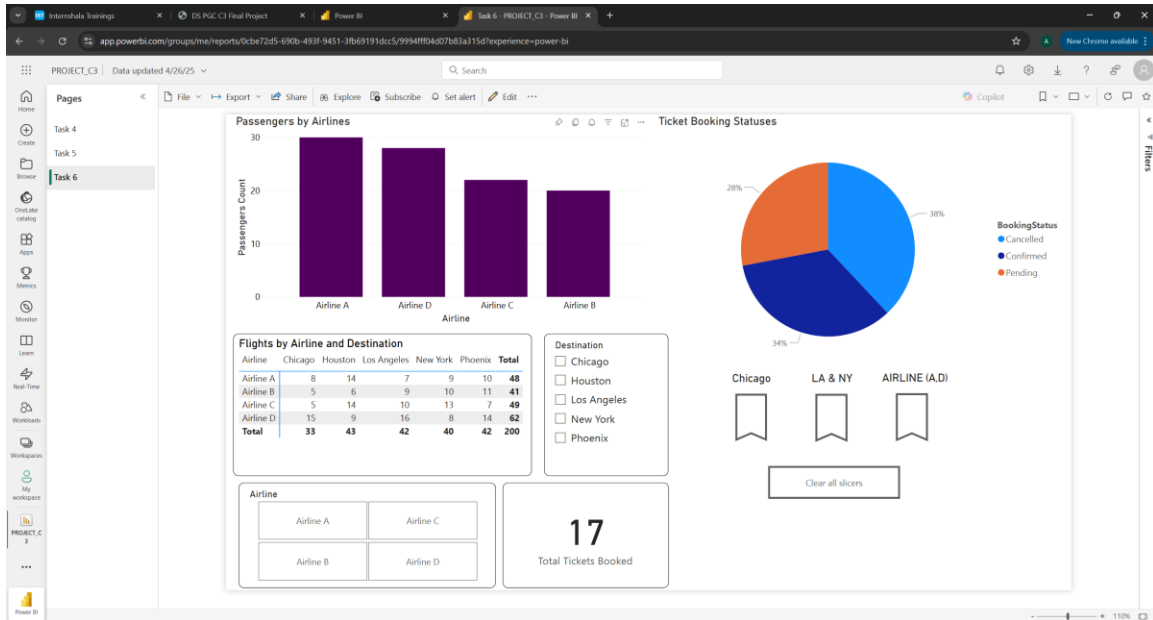


Insight: Visualizing the data revealed hidden trends, and adding interactivity made it easier for users to explore specific details on their own.

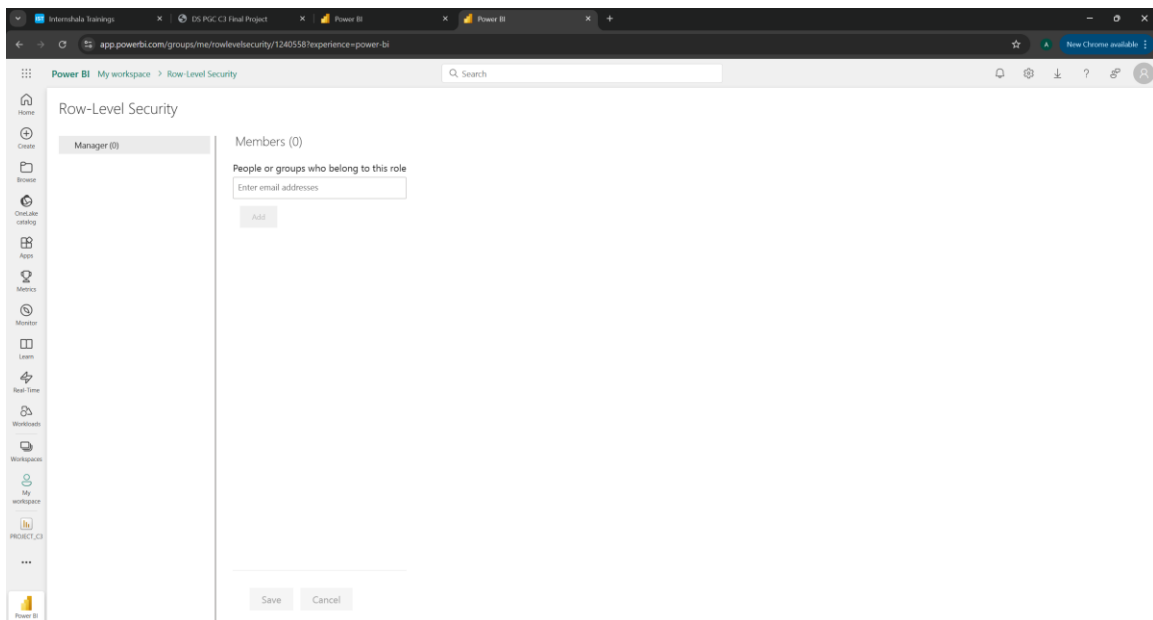
Task 6: Final Dashboard and Power BI Service

I designed a final comprehensive dashboard that compiled all major visuals and insights cohesively.

I also configured Row-Level Security (RLS) for Airline A and scheduled a daily refresh at 5 PM to keep the dashboard data current.



Enabled RLS



Scheduled refresh

The screenshot displays the Microsoft Power BI 'My workspace' interface. On the left is a navigation pane with icons for Home, Create, Browse, OneLake catalog, Apps, Metrics, Monitor, Learn, Real-Time, Workloads, Workspaces, My workspace, and PROJECT_C3. The main content area has a search bar at the top. Below it, a yellow banner states 'Scheduled refresh has been disabled. [Show details](#)'. The 'Refresh history' section is collapsed. The 'Semantic model description' section contains a text box for describing the model's contents, with a '500 characters left' indicator. Below this are 'Apply' and 'Discard' buttons. The 'Gateway and cloud connections' section is also collapsed. The 'Data source credentials' section lists three data sources: 'Flight_Information - flight_information.csv', 'Passenger_Information - passenger_information.csv', and 'Ticket_Information - ticket_information.csv', each with 'Edit credentials' and 'Show in lineage view' links. The 'Parameters' section is collapsed. The 'Refresh' section is expanded, showing 'Time zone' set to '(UTC) Coordinated Universal Time' and a note about time zone configuration. Below this, 'Configure a refresh schedule' is set to 'On', 'Refresh frequency' is 'Daily', and 'Time' is set to '5:00 AM'.

Insight: Publishing the dashboard with security and automatic refresh enhanced its professionalism and made it ready for real-world enterprise use.

[End of Report]

[Ansh kumar]

Video Explanation:

https://drive.google.com/drive/folders/1RAXY0uvcT-ATD2552MjY-2vJxYbo3dLP?usp=drive_link